84. Match the following columns and select the **correct** option.

	Column - I		**Column - II**
(a)	Floating Ribs	(i)	Located between second and seventh ribs
(b)	Acromion	(ii)	Head of the Humerus
(c)	Scapula	(iii)	Clavicle
(d)	Glenoid cavity	(iv)	Do not connect with the sternum

	(a)	**(b)**	**(c)**	**(d)**
(1)	(ii)	(iv)	(i)	(iii)
(2)	(i)	(iii)	(ii)	(iv)
(3)	(iii)	(ii)	(iv)	(i)
(4)	(iv)	(iii)	(i)	(ii)

85. The number of substrate level phosphorylations in one turn of citric acid cycle is :

(1) Zero
(2) One
(3) Two
(4) Three

86. Dissolution of the synaptonemal complex occurs during :

(1) Pachytene
(2) Zygotene
(3) Diplotene
(4) Leptotene

87. Bilaterally symmetrical and acoelomate animals are exemplified by :

(1) Ctenophora
(2) Platyhelminthes
(3) Aschelminthes
(4) Annelida

88. The body of the ovule is fused within the funicle at :

(1) Hilum
(2) Micropyle
(3) Nucellus
(4) Chalaza

89. Goblet cells of alimentary canal are modified from :

(1) Squamous epithelial cells
(2) Columnar epithelial cells
(3) Chondrocytes
(4) Compound epithelial cells

90. Snow-blindness in Antarctic region is due to :

(1) Freezing of fluids in the eye by low temperature
(2) Inflammation of cornea due to high dose of UV-B radiation
(3) High reflection of light from snow
(4) Damage to retina caused by infra-red rays

91. Identify a molecule which does **not** exist.

(1) He_2
(2) Li_2
(3) C_2
(4) O_2

92. Find out the solubility of $Ni(OH)_2$ in 0.1 M NaOH. Given that the ionic product of $Ni(OH)_2$ is 2×10^{-15}.

(1) 2×10^{-13} M
(2) 2×10^{-8} M
(3) 1×10^{-13} M
(4) 1×10^{8} M

93. Identify the **correct** statements from the following :

(a) $CO_2(g)$ is used as refrigerant for ice-cream and frozen food.
(b) The structure of C_{60} contains twelve six carbon rings and twenty five carbon rings.
(c) ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
(d) CO is colorless and odourless gas.

(1) (a), (b) and (c) only
(2) (a) and (c) only
(3) (b) and (c) only
(4) (c) and (d) only

94. Hydrolysis of sucrose is given by the following reaction.

$$\text{Sucrose} + H_2O \rightleftharpoons \text{Glucose} + \text{Fructose}$$

If the equilibrium constant (K_c) is 2×10^{13} at 300 K, the value of $\Delta_r G^{\ominus}$ at the same temperature will be :

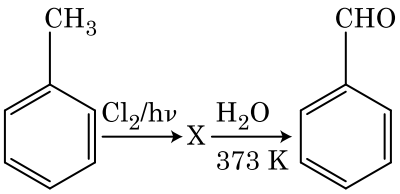
(1) $-8.314\ \text{J mol}^{-1}\text{K}^{-1} \times 300\ \text{K} \times \ln(2 \times 10^{13})$

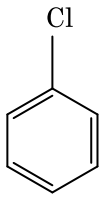
(2) $8.314\ \text{J mol}^{-1}\text{K}^{-1} \times 300\ \text{K} \times \ln(2 \times 10^{13})$

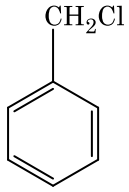
(3) $8.314\ \text{J mol}^{-1}\text{K}^{-1} \times 300\ \text{K} \times \ln(3 \times 10^{13})$

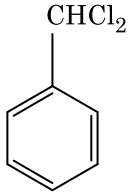
(4) $-8.314\ \text{J mol}^{-1}\text{K}^{-1} \times 300\ \text{K} \times \ln(4 \times 10^{13})$

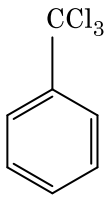
95. Identify compound X in the following sequence of reactions :



(1) 

(2) 

(3) 

(4) 

96. Identify the **incorrect** match.

	Name		**IUPAC Official Name**
(a)	Unnilunium	(i)	Mendelevium
(b)	Unniltrium	(ii)	Lawrencium
(c)	Unnilhexium	(iii)	Seaborgium
(d)	Ununnunnium	(iv)	Darmstadtium

(1) (a), (i)

(2) (b), (ii)

(3) (c), (iii)

(4) (d), (iv)

97. An element has a body centered cubic (bcc) structure with a cell edge of 288 pm. The atomic radius is :

(1) $\frac{\sqrt{3}}{4} \times 288\ \text{pm}$

(2) $\frac{\sqrt{2}}{4} \times 288\ \text{pm}$

(3) $\frac{4}{\sqrt{3}} \times 288\ \text{pm}$

(4) $\frac{4}{\sqrt{2}} \times 288\ \text{pm}$

98. Which of the following set of molecules will have zero dipole moment ?

(1) Ammonia, beryllium difluoride, water, 1,4-dichlorobenzene

(2) Boron trifluoride, hydrogen fluoride, carbon dioxide, 1,3-dichlorobenzene

(3) Nitrogen trifluoride, beryllium difluoride, water, 1,3-dichlorobenzene

(4) Boron trifluoride, beryllium difluoride, carbon dioxide, 1,4-dichlorobenzene

99. On electrolysis of dil.sulphuric acid using Platinum (Pt) electrode, the product obtained at anode will be :

(1) Hydrogen gas

(2) Oxygen gas

(3) H_2S gas

(4) SO_2 gas

100. Reaction between acetone and methylmagnesium chloride followed by hydrolysis will give :

(1) Isopropyl alcohol

(2) Sec. butyl alcohol

(3) Tert. butyl alcohol

(4) Isobutyl alcohol

101. Which of the following oxoacid of sulphur has $-O-O-$ linkage ?

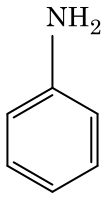
(1) H_2SO_3, sulphurous acid

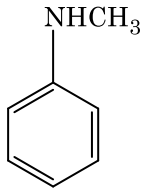
(2) H_2SO_4, sulphuric acid

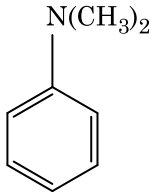
(3) $H_2S_2O_8$, peroxodisulphuric acid

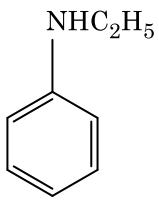
(4) $H_2S_2O_7$, pyrosulphuric acid

102. Which of the following amine will give the carbylamine test ?

(1) 

(2) 

(3) 

(4) 

103. The calculated spin only magnetic moment of Cr^{2+} ion is :

(1) 3.87 BM

(2) 4.90 BM

(3) 5.92 BM

(4) 2.84 BM

104. The correct option for free expansion of an ideal gas under adiabatic condition is :

(1) $q = 0$, $\Delta T = 0$ and $w = 0$

(2) $q = 0$, $\Delta T < 0$ and $w > 0$

(3) $q < 0$, $\Delta T = 0$ and $w = 0$

(4) $q > 0$, $\Delta T > 0$ and $w > 0$

105. The freezing point depression constant (K_f) of benzene is 5.12 K kg mol^{-1}. The freezing point depression for the solution of molality 0.078 m containing a non-electrolyte solute in benzene is (rounded off upto two decimal places) :

(1) 0.20 K

(2) 0.80 K

(3) 0.40 K

(4) 0.60 K

106. The number of Faradays(F) required to produce 20 g of calcium from molten $CaCl_2$ (Atomic mass of Ca = 40 g mol^{-1}) is :

(1) 1

(2) 2

(3) 3

(4) 4

107. Reaction between benzaldehyde and acetophenone in presence of dilute NaOH is known as :

(1) Aldol condensation

(2) Cannizzaro's reaction

(3) Cross Cannizzaro's reaction

(4) Cross Aldol condensation

108. Paper chromatography is an example of :

(1) Adsorption chromatography

(2) Partition chromatography

(3) Thin layer chromatography

(4) Column chromatography

109. An increase in the concentration of the reactants of a reaction leads to change in :

(1) activation energy

(2) heat of reaction

(3) threshold energy

(4) collision frequency

110. A mixture of N_2 and Ar gases in a cylinder contains 7 g of N_2 and 8 g of Ar. If the total pressure of the mixture of the gases in the cylinder is 27 bar, the partial pressure of N_2 is :

[Use atomic masses (in g mol^{-1}) : N = 14, Ar = 40]

(1) 9 bar

(2) 12 bar

(3) 15 bar

(4) 18 bar

111. Identify the **correct** statement from the following :

(1) Wrought iron is impure iron with 4% carbon.

(2) Blister copper has blistered appearance due to evolution of CO_2.

(3) Vapour phase refining is carried out for Nickel by Van Arkel method.

(4) Pig iron can be moulded into a variety of shapes.

112. A tertiary butyl carbocation is more stable than a secondary butyl carbocation because of which of the following ?

(1) $-$I effect of $-CH_3$ groups

(2) $+$R effect of $-CH_3$ groups

(3) $-$R effect of $-CH_3$ groups

(4) Hyperconjugation

113. Which of the following is a cationic detergent ?

(1) Sodium lauryl sulphate

(2) Sodium stearate

(3) Cetyltrimethyl ammonium bromide

(4) Sodium dodecylbenzene sulphonate

114. Elimination reaction of 2-Bromo-pentane to form pent-2-ene is :

(a) β-Elimination reaction

(b) Follows Zaitsev rule

(c) Dehydrohalogenation reaction

(d) Dehydration reaction

(1) (a), (b), (c)

(2) (a), (c), (d)

(3) (b), (c), (d)

(4) (a), (b), (d)

115. The mixture which shows positive deviation from Raoult's law is :

(1) Ethanol + Acetone

(2) Benzene + Toluene

(3) Acetone + Chloroform

(4) Chloroethane + Bromoethane

116. Which of the following is the **correct** order of increasing field strength of ligands to form coordination compounds ?

(1) $SCN^- < F^- < C_2O_4^{2-} < CN^-$

(2) $SCN^- < F^- < CN^- < C_2O_4^{2-}$

(3) $F^- < SCN^- < C_2O_4^{2-} < CN^-$

(4) $CN^- < C_2O_4^{2-} < SCN^- < F^-$

117. Which of the following is a basic amino acid ?

(1) Serine

(2) Alanine

(3) Tyrosine

(4) Lysine

118. HCl was passed through a solution of $CaCl_2$, $MgCl_2$ and NaCl. Which of the following compound(s) crystallise(s) ?

(1) Both $MgCl_2$ and $CaCl_2$

(2) Only NaCl

(3) Only $MgCl_2$

(4) NaCl, $MgCl_2$ and $CaCl_2$

119. Which of the following is a natural polymer ?

(1) *cis*-1,4-polyisoprene

(2) poly (Butadiene-styrene)

(3) polybutadiene

(4) poly (Butadiene-acrylonitrile)

120. Which of the following is **not** correct about carbon monoxide ?

(1) It forms carboxyhaemoglobin.

(2) It reduces oxygen carrying ability of blood.

(3) The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.

(4) It is produced due to incomplete combustion.

121. Sucrose on hydrolysis gives :

(1) β-D-Glucose + α-D-Fructose

(2) α-D-Glucose + β-D-Glucose

(3) α-D-Glucose + β-D-Fructose

(4) α-D-Fructose + β-D-Fructose

122. The following metal ion activates many enzymes, participates in the oxidation of glucose to produce ATP and with Na, is responsible for the transmission of nerve signals.

(1) Iron

(2) Copper

(3) Calcium

(4) Potassium

123. Which one of the followings has maximum number of atoms ?

(1) 1 g of Ag(s) [Atomic mass of Ag = 108]

(2) 1 g of Mg(s) [Atomic mass of Mg = 24]

(3) 1 g of O_2(g) [Atomic mass of O = 16]

(4) 1 g of Li(s) [Atomic mass of Li = 7]

124. The number of protons, neutrons and electrons in $^{175}_{71}Lu$, respectively, are :

(1) 71, 104 and 71

(2) 104, 71 and 71

(3) 71, 71 and 104

(4) 175, 104 and 71

125. What is the change in oxidation number of carbon in the following reaction ?

$CH_4(g) + 4Cl_2(g) \rightarrow CCl_4(l) + 4HCl(g)$

(1) +4 to +4

(2) 0 to +4

(3) −4 to +4

(4) 0 to −4

126. Identify the **incorrect** statement.

(1) $Cr^{2+}(d^4)$ is a stronger reducing agent than $Fe^{2+}(d^6)$ in water.

(2) The transition metals and their compounds are known for their catalytic activity due to their ability to adopt multiple oxidation states and to form complexes.

(3) Interstitial compounds are those that are formed when small atoms like H, C or N are trapped inside the crystal lattices of metals.

(4) The oxidation states of chromium in CrO_4^{2-} and $Cr_2O_7^{2-}$ are not the same.

127. For the reaction, $2Cl(g) \rightarrow Cl_2(g)$, the **correct** option is :

(1) $\Delta_r H > 0$ and $\Delta_r S > 0$

(2) $\Delta_r H > 0$ and $\Delta_r S < 0$

(3) $\Delta_r H < 0$ and $\Delta_r S > 0$

(4) $\Delta_r H < 0$ and $\Delta_r S < 0$

128. Measuring Zeta potential is useful in determining which property of colloidal solution ?

(1) Viscosity

(2) Solubility

(3) Stability of the colloidal particles

(4) Size of the colloidal particles

129. Urea reacts with water to form **A** which will decompose to form **B**. **B** when passed through Cu^{2+} (aq), deep blue colour solution **C** is formed. What is the formula of **C** from the following ?

(1) $CuSO_4$

(2) $[Cu(NH_3)_4]^{2+}$

(3) $Cu(OH)_2$

(4) $CuCO_3 \cdot Cu(OH)_2$

130. Match the following and identify the **correct** option.

(a)	$CO(g) + H_2(g)$	(i)	$Mg(HCO_3)_2 + Ca(HCO_3)_2$
(b)	Temporary hardness of water	(ii)	An electron deficient hydride
(c)	B_2H_6	(iii)	Synthesis gas
(d)	H_2O_2	(iv)	Non-planar structure

	(a)	**(b)**	**(c)**	**(d)**
(1)	(iii)	(i)	(ii)	(iv)
(2)	(iii)	(ii)	(i)	(iv)
(3)	(iii)	(iv)	(ii)	(i)
(4)	(i)	(iii)	(ii)	(iv)

131. Match the following :

	Oxide		**Nature**
(a)	CO	(i)	Basic
(b)	BaO	(ii)	Neutral
(c)	Al_2O_3	(iii)	Acidic
(d)	Cl_2O_7	(iv)	Amphoteric

Which of the following is **correct** option ?

	(a)	**(b)**	**(c)**	**(d)**
(1)	(i)	(ii)	(iii)	(iv)
(2)	(ii)	(i)	(iv)	(iii)
(3)	(iii)	(iv)	(i)	(ii)
(4)	(iv)	(iii)	(ii)	(i)

132. The rate constant for a first order reaction is $4.606 \times 10^{-3}\ s^{-1}$. The time required to reduce 2.0 g of the reactant to 0.2 g is :

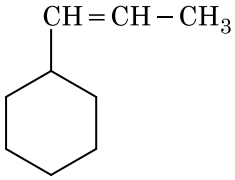
(1) 100 s

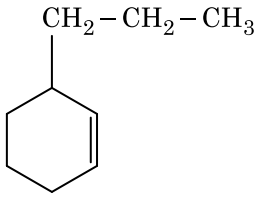
(2) 200 s

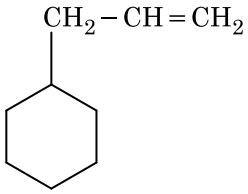
(3) 500 s

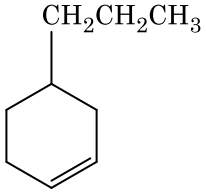
(4) 1000 s

133. An alkene on ozonolysis gives methanal as one of the product. Its structure is :

(1) 

(2) 

(3) 

(4) 

134. Which of the following alkane cannot be made in good yield by Wurtz reaction ?

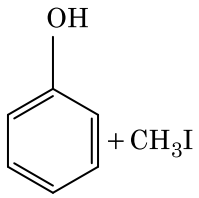
(1) n-Hexane

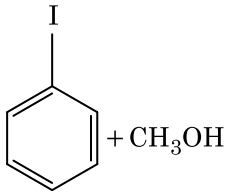
(2) 2,3-Dimethylbutane

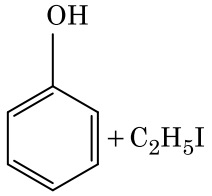
(3) n-Heptane

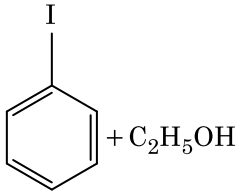
(4) n-Butane

135. Anisole on cleavage with HI gives :

(1) 

(2) 

(3) 

(4) 

136. For which one of the following, Bohr model is **not** valid ?

(1) Hydrogen atom

(2) Singly ionised helium atom (He^+)

(3) Deuteron atom

(4) Singly ionised neon atom (Ne^+)

137. The ratio of contributions made by the electric field and magnetic field components to the intensity of an electromagnetic wave is : (c = speed of electromagnetic waves)

(1) c : 1

(2) 1 : 1

(3) 1 : c

(4) $1 : c^2$

138. The Brewsters angle i_b for an interface should be :

(1) $0° < i_b < 30°$

(2) $30° < i_b < 45°$

(3) $45° < i_b < 90°$

(4) $i_b = 90°$

139. A cylinder contains hydrogen gas at pressure of 249 kPa and temperature 27°C.

Its density is : ($R = 8.3\ J\ mol^{-1}\ K^{-1}$)

(1) $0.5\ kg/m^3$

(2) $0.2\ kg/m^3$

(3) $0.1\ kg/m^3$

(4) $0.02\ kg/m^3$

140. A ray is incident at an angle of incidence i on one surface of a small angle prism (with angle of prism A) and emerges normally from the opposite surface. If the refractive index of the material of the prism is μ, then the angle of incidence is nearly equal to :

(1) $\frac{A}{2\mu}$

(2) $\frac{2A}{\mu}$

(3) μA

(4) $\frac{\mu A}{2}$

141. Two cylinders A and B of equal capacity are connected to each other via a stop cock. A contains an ideal gas at standard temperature and pressure. B is completely evacuated. The entire system is thermally insulated. The stop cock is suddenly opened. The process is :

(1) isothermal

(2) adiabatic

(3) isochoric

(4) isobaric

142. The energy equivalent of 0.5 g of a substance is :

(1) $4.5 \times 10^{16}\ J$

(2) $4.5 \times 10^{13}\ J$

(3) $1.5 \times 10^{13}\ J$

(4) $0.5 \times 10^{13}\ J$

143. A body weighs 72 N on the surface of the earth. What is the gravitational force on it, at a height equal to half the radius of the earth ?

(1) 48 N

(2) 32 N

(3) 30 N

(4) 24 N